



Healthy Skepticism Day 3

Where the lines get drawn, and who draws them



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The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

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The Five Faults

1

Relative reporting

Day 1

What does a 36% improvement from taking a drug really mean?

2

Probability reversal

Day 2

What does a positive COVID test really mean?

3

Arbitrary categories

Day 3

How concerned should you be if your systolic BP goes from 119 to 120?

4

The Bernoulli fallacy

Day 3

Why does the published medical advice keep changing?

5

The causal leap

Day 4

Does bird watching prevent dementia?

Once you see them, you can't unsee them.



Today's Two Faults

3

Arbitrary categories

Blood pressure, body weight, and where the lines fall

4

The Bernoulli fallacy

The 0.05 line, and what a study can and cannot tell you



FAULT 3

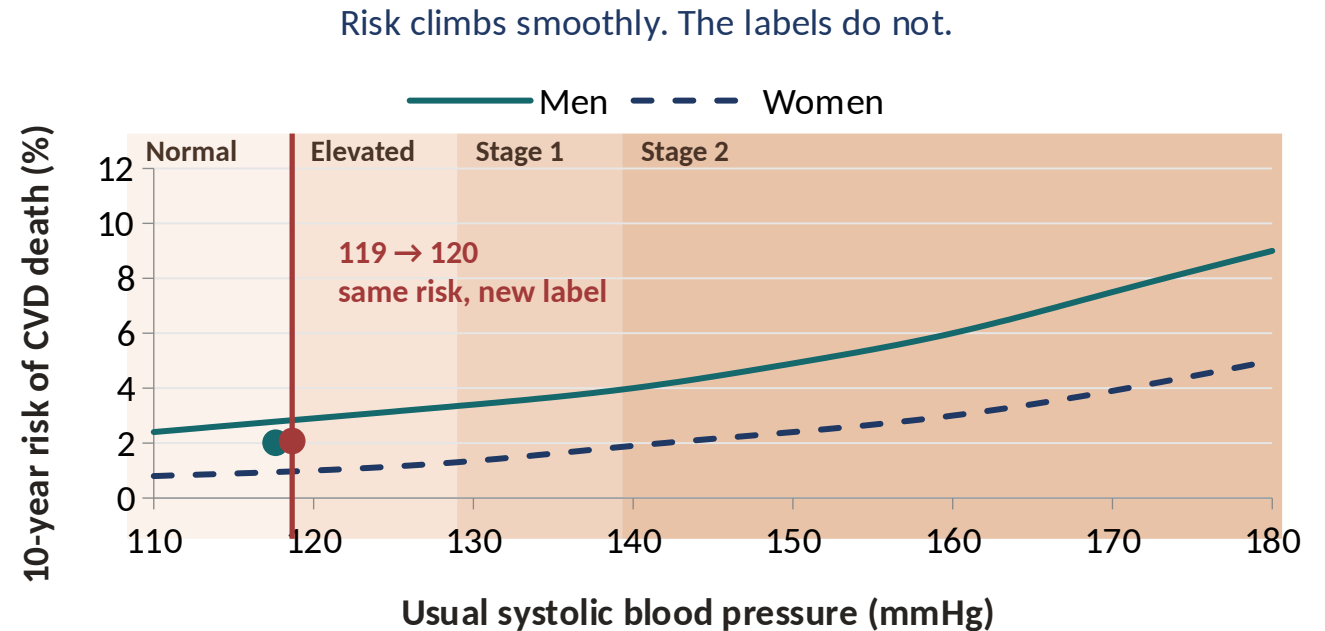
Arbitrary Categories

Read the curve, not the cutoff



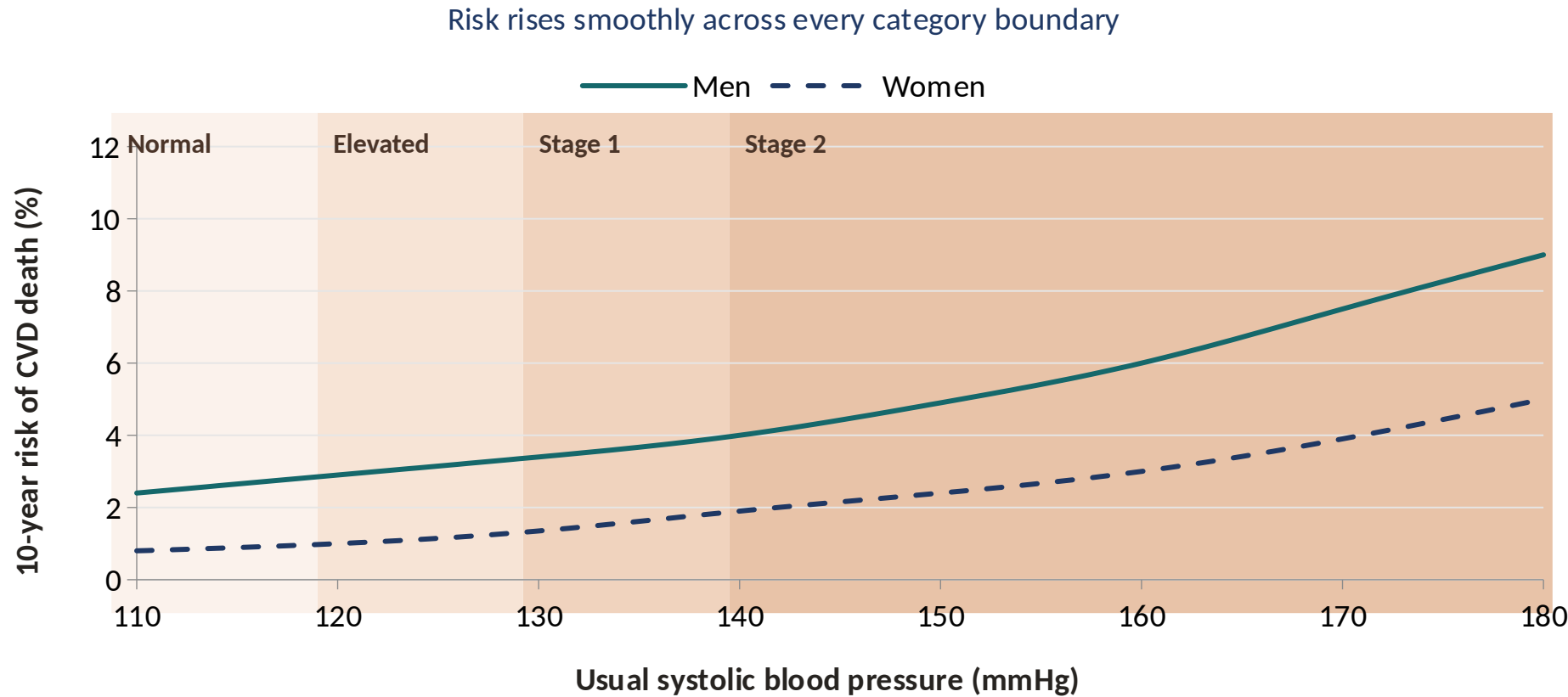
Where Does “High” Begin?

- Systolic this year: 120. Last year: 119.
- Nothing about you changed. The label did.



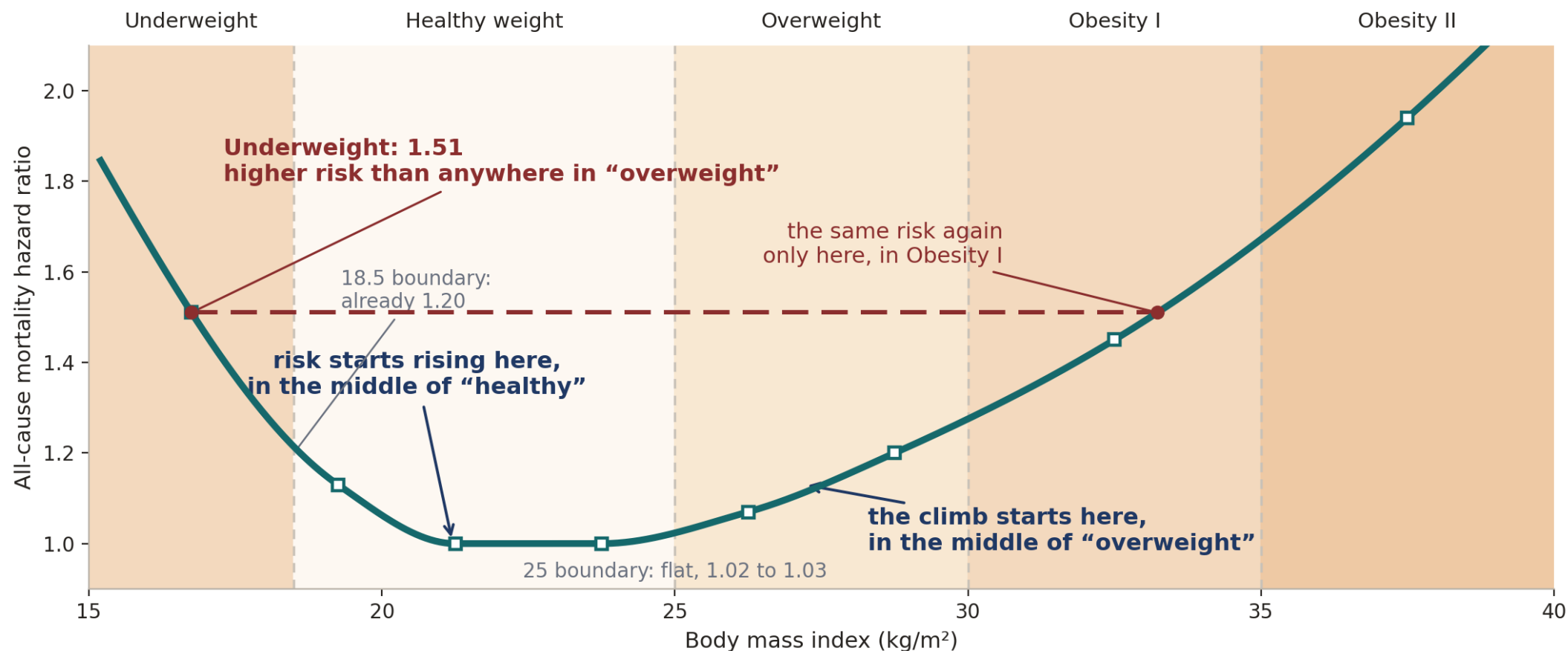
One mmHg moved you across a line. It did not move your risk.

Systolic Categories Don't Mean Much



SCORE risk charts, low-risk regions: Conroy RM et al., Eur Heart J 2003;24:987-1003. Category boundaries: ACC/AHA 2017 hypertension guideline.

BMI Categories Are Worse



The risk transitions sit in the middle of the categories; the boundaries sit where nothing happens.

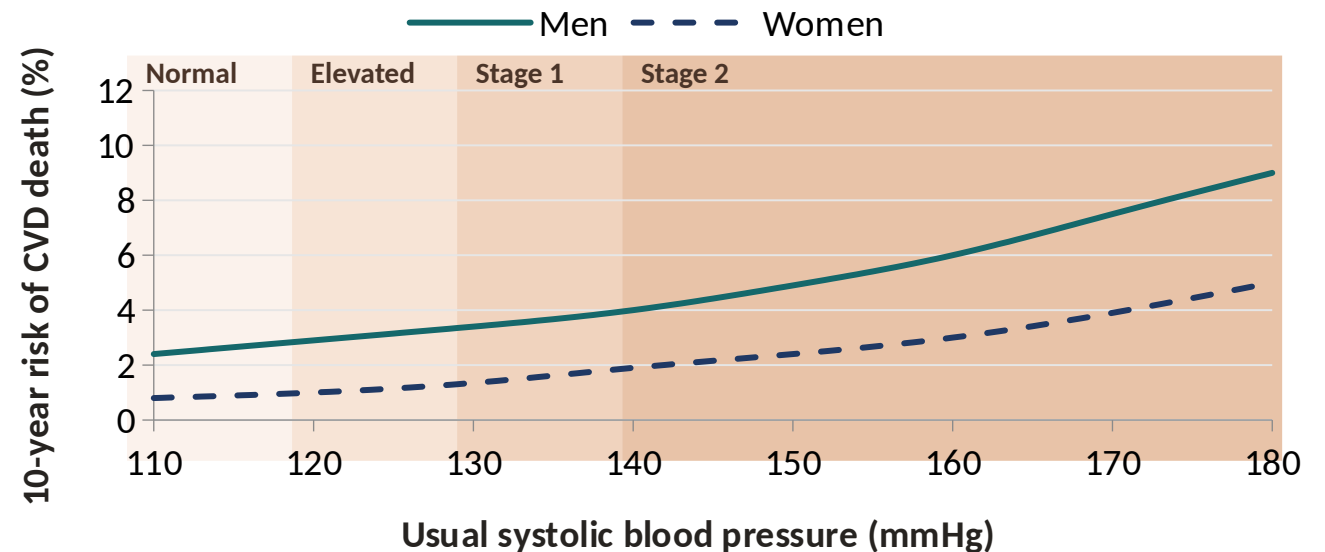
Global BMI Mortality Collaboration, Lancet 2016;388:776-786. All-cause mortality hazard ratios relative to BMI 22.5-25, never-smokers; band-midpoint values, shape-preserving interpolation.



So Why Have Categories at All?

- A box is a decision rule: triage in three seconds.
- The trouble: the box gets read as a fact about your body.
- A box cannot say how much risk you carry.

The risk curve does not know where the categories are



Ask where you sit on the curve, not which side of the line.



FAULT 4

The Bernoulli Fallacy

Why medical advice keeps changing

Published Papers with Findings That Could be Repeated

Only 6 of 53 landmark cancer findings were confirmed on rerun.

Just 35 of 97 psychology findings stayed significant on rerun.

This is the Irreproducibility Crisis

An Important Aside: $P(A | B)$ is The Core Element of Inference

- It is called 'conditional probability'
- It means the probability of A if B is true
- Sometimes shortened to the probability of A, given B
- Depending on the circumstances, B may be observed, believed, or imagined for argument's sake
- **Examples:**
 - Courtrooms: $P(\text{Criminal guilt} \mid \text{evidence})$
 - Gambling: $P(\text{Rolling 7 with 2 dice} \mid \text{Dice are fair})$
 - Science: $P(\text{Tested hypothesis} \mid \text{data})$
 - Causality: $P(\text{drowning} \mid \text{warm weather})$

For the course, you only need to be aware of the concept

Sometimes We Want to Reverse Conditional Probability

Day 2 had 2 examples

- As we saw in Day 2, we can reverse $P(A|B)$ to $P(B|A)$ if we also have $P(B)$
- $P(B)$ is sometimes called the 'prevalence' or 'prior' depending on the circumstances
- For example, in the mammogram example
 - $P(\text{called back} | \text{cancer}) = 87\%$
 - $P(\text{cancer}) = 0.6\%$ (6 in 1000)
 - We computed using ratios $P(\text{cancer} | \text{called back}) = 4.4\%$

This sort of calculation, reversing using the prior, is called 'Bayesian Analysis'

The Fallacy Stems From Frequentist Reasoning

- They want to know if a hypothesis is true.
- Rather than compute $P(\text{hypothesis} \mid \text{data})$, they compute $p = P(\text{data} \mid \text{hypothesis is false})$
- If p is low enough, they say the observations are 'statistically significant' and submit the result for publication.
- In medicine and the social sciences, the significance criterion is $p < .05$: if the hypothesis were false, the data would arise only 5% of the time.
 - Some characterize this if the hypothesis were false, the data is very surprising, so can treat the hypothesis as true

This criterion means one would expect about 1 in 20 published experiments to be not reproducible.

The Drawbacks

- 1 **The significance setting is arbitrary, and often far too high.** There is no formula or theorem that defines 'significance' .
- 2 **Significance is not informative.** Knowing $P(\text{hypothesis} \mid \text{data})$ enables risk-based decision support, like gambling; the p-value does not.
- 3 **It invites p-hacking.** Searching subsets of the data for those that meet the criterion by luck alone.

Together these feed the replication crisis

Medical and Social Science Sets the p-value Far Too High for Significance

Medicine cannot wait for five-sigma data: patients are scarce, trials costly.

WHAT PHYSICS DEMANDS

$$p < 0.000000287$$

Five sigma: one chance in 3.5 million. The Higgs waited for that bar.

WHAT MEDICINE AND SOCIAL SCIENCE ACCEPT

$$p < 0.05$$

One chance in twenty. A no-effect test clears it 1 time in 20, by luck.

1 in 20 is the rate per test of nothing. The share of false findings is a reversal question: how many tested ideas were true? The prevalence, yet again.

So Why Do They Adopt Frequentism?

- **Sometimes there is no prior:**
 - Industrial quality control
 - Census survey sampling
 - Particle physics (which sets the criterion at $p < 0.000000287$)
 - Not using a prior is considered more objective and honest by its proponents.
- **As described in the book**, a more rigorous standard disrupts the academic publication world.

Evidence of p-hacking

Only 6 of 53 (11%) landmark cancer findings were confirmed on rerun.

Just 35 of 97 (36%) psychology findings stayed significant on rerun.

These are much lower than chance (95%)

Evidence for ‘p-hacking’, aka ‘data torturing’

The observed failure rate is far above the honest floor.

1 in 20

the failure rate honest testing would produce on its own, with no torturing at all.

47 of 53

landmark cancer findings that failed to repeat. Psychology: about 2 in 3.

The failure rate is too high for luck alone.

The Baby Aspirin Recommendation Change

Trials stopped early tend to stop on a lucky high. On the retests, p never crossed the line again.

1988

Physicians' Health Study: aspirin cut first heart attacks 44%, $p < 0.00001$. Stopped early for benefit. Decades of daily-aspirin advice followed.

2018

The retests, ARRIVE and ASPREE: no net benefit (ARRIVE $p = 0.60$), more major bleeding. USPSTF 2022: do not start after 60.

The advice changed because the finding would not repeat, and the harm outweighed the benefit

Autism: From a p-Value to Policy

31

autism cases, parent-reported, not
medical records

3x

the reported odds, in boys only, not
the full sample

$p < 0.05$

so it was written up as a finding

- 1 **One study:** 31 parent-reported cases.
- 2 **One number:** p below 0.05.
- 3 **One citation:** the CDC's Autism and Vaccines page.
- 4 **One committee:** ACIP, twice, days before a vote.
- 5 **One policy:** universal newborn hepatitis B vaccination, ended.

Never replicated. In May 2026 the journal retracted the paper, months after the vote.



What We've Learned

Two faults, one shape.

1

1 · Arbitrary categories

The trap: A line drawn across a continuous measurement is read as a fact about your body.

Ask: What is my risk with my measurement?

2

2 · The Bernoulli fallacy

The trap: p below 0.05 is read as proof the effect is real has led to unreliable medical literature

Ask: Have these studies been replicated?



Questions, and a conversation

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Backup Slides



A Line You Have Already Met

- Arbitrary categories came from clinical medicine: BMI 25, systolic 130.
- Statistics draws the same kind of line on itself. A study is "significant" if p falls below 0.05, and not otherwise.
- p is a continuous quantity. 0.05 is a convention, not a discovery. R. A. Fisher drew it in 1925 because it was convenient.
- Everything Fault 3 taught about cutoffs applies to the line the research itself is graded on.



Too Few Cases, Read as No Effect

- [Placeholder: ASCOT-LLA's women.]
- Only 36 heart events among women in the whole trial. The label calls the result inconclusive.
- Correspondents read that as evidence the drug does not help women.
- The investigators replied that 36 events cannot show much, and that absent evidence of a different effect, the whole-trial result is the best estimate for women too.
- Source: Sever PS, et al. Lancet 2003;361:1986.



Sixty Side Effects at Once

- [Placeholder: the 2026 statin label meta-analysis.]
- More than sixty labelled side effects tested, with a strict bar to avoid finding things by luck.
- Six survived. The conclusion drawn was that the labels should be trimmed.
- A strict bar also means real effects get missed, and that rate was never reported.
- Source: Cholesterol Treatment Trialists' Collaboration, Lancet 2026;407:689-703.